

BRIDGEPULSE · DESIGN DECISION RECORD

BridgePulse far gauge cable route between the girders

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	BRP-DDR-003	Version	0.1
Status	Draft	Date	2026-09-27
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/bridgepulse		
Source	docs/decisions/0003-far-gauge-cable-route.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-27	Amish Chadha	Record Amish's decision to move the far gauge cable's vertical runs inboard of the bottom flanges in cad/src/model.py (REVIEW.md, session 2026-09-26, item 4)

0003: Far gauge cable route between the girders

- **Date:** 2026-09-27
- **Status:** accepted. Decided by Amish on 2026-09-27.

■ Context

While routing cables for the product appearance model (`docs/REVIEW.md` , session 2026-09-26, item 4), the review found a modelling error in `cad/src/model.py` . The far gauge cable, which runs from the north girder's gauge across to the hub on the south girder, rose and fell between the girders at $|y| = 522$ mm. The inner edges of the bottom flanges are at $|y| = 515$ mm (girder centerline 600 mm less half the 170 mm flange), so the modelled 8 mm diameter cable ran 11 mm into each bottom flange. The error was in the sign of two terms in `inner_n` , not in the design intent, which is a cable clipped under each flange and rising clear of it.

■ Options considered

1. Change `inner_n` in `cad/src/model.py` from `gy - bf / 2 + r + 3` to `gy - bf / 2 - r - 3` , placing the vertical runs one cable radius plus 3 mm inboard of the inner flange edges (recommended in the review).
2. Leave `model.py` as it is and correct only the appearance model (the state after 2026-09-26).

■ Decision

Option 1. Decided by Amish on 2026-09-27. Amish's instruction: "resolve the challenges for ConePro, BridgePulse, Grainguard and WellSense."

The formula now lives in one function, `far_cable_inner()` in `cad/src/model.py` , which `build_parts()` uses for `inner_n` and `cad/src/product_model.py` imports for its own far gauge cable, so the two models cannot drift apart again.

■ Consequences

- The far gauge cable's vertical runs are at $|y| = 508$ mm. Its surface is 3 mm inboard of each inner flange edge; the solid overlap between the cables and the girders in the model is zero (was 1,277 mm³ for the far cable alone), and the closest approach anywhere is the intended 2 mm under the flange where the cable is clipped.
- Every other dimension and interface is unchanged. No change to the BOM, the budget, the requirements or the calculations: the calc note does not use the cable route, and the clearance below the soffit (R9) is set by the cable runs under the flanges, which did not move.
- Regenerated from `model.py` : `cad/step/*.step` , `cad/stl/*.stl` , the concept media in `media/` , and drawing BRP-DWG-001, now Rev P3.
- The appearance model already used $|y| = 508$ mm, so the photoreal renders (`media/render-*.png`) do not change.
- Other open items from the 2026-09-26 review (lid window, parts left out, cable routes shortened and moved, probe sheath size) remain "Proposed, awaiting Amish".
- TRL stays at 3; TRL 4 remains on hold.